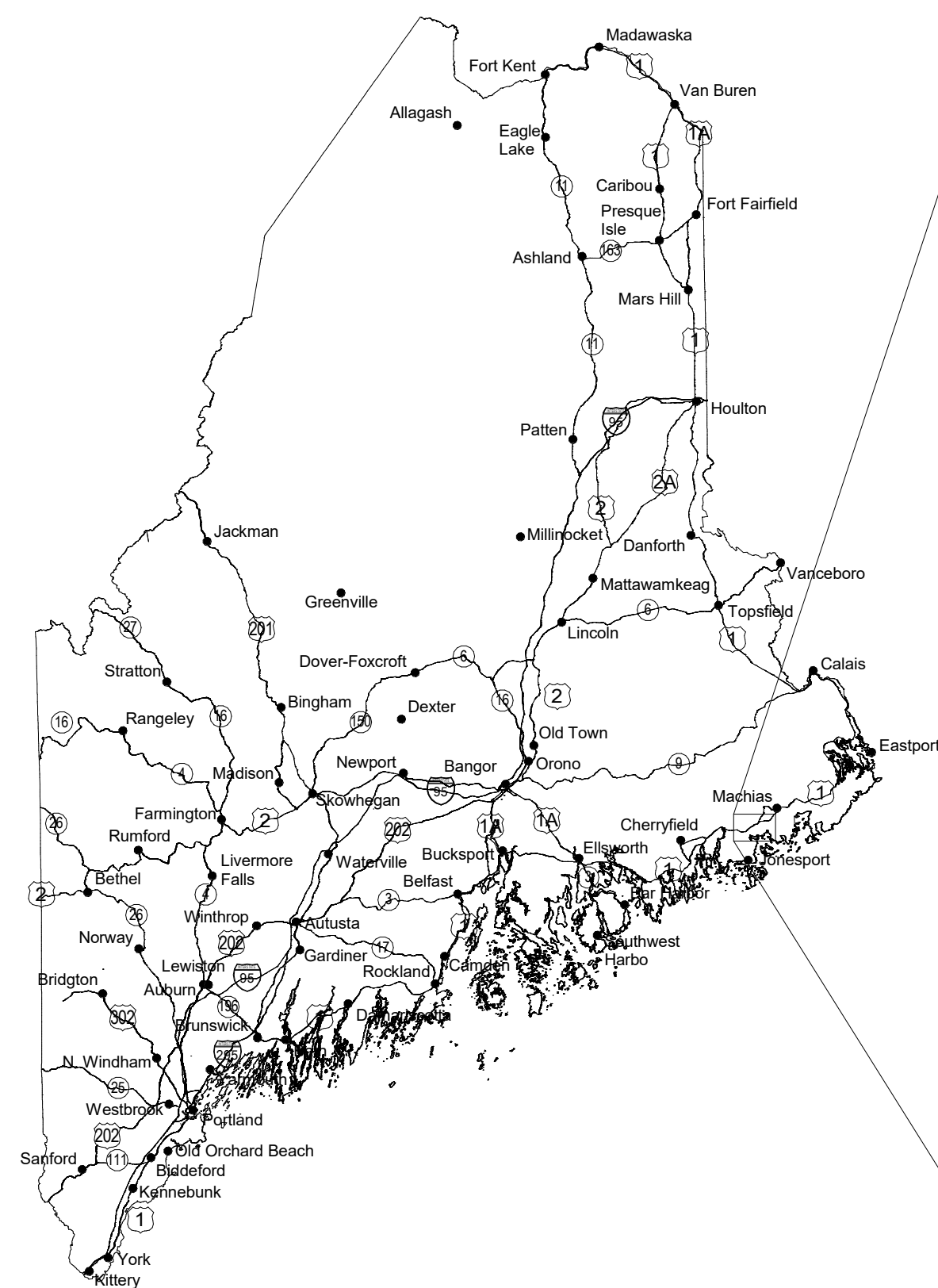


MAINE IF&W HEAT PUMP & ELECTRICAL UPGRADES

JONESBORO, MAINE



DRAWING LIST				
SHEET NUMBER	SHEET TITLE	REVISION	DATE	DESCRIPTION
M-001	MECHANICAL NOTES, SYMBOLS & ABBREVIATIONS			
M-101	MECHANICAL BASEMENT PLAN			
M-102	MECHANICAL FIRST FLOOR PLAN			
M-501	MECHANICAL DETAILS			
M-502	MECHANICAL DETAILS			
M-503	MECHANICAL DETAILS			
M-601	MECHANICAL SCHEDULES			
E-001	ELECTRICAL ABBREVIATIONS, NOTES, DETAILS & SCHEDULES			
E-100	ELECTRICAL SITE PLAN AND ONE-LINE DIAGRAMS			
E-101	ELECTRICAL POWER PLANS			

ISSUED FOR BID

2026.04.23



HALEY WARD

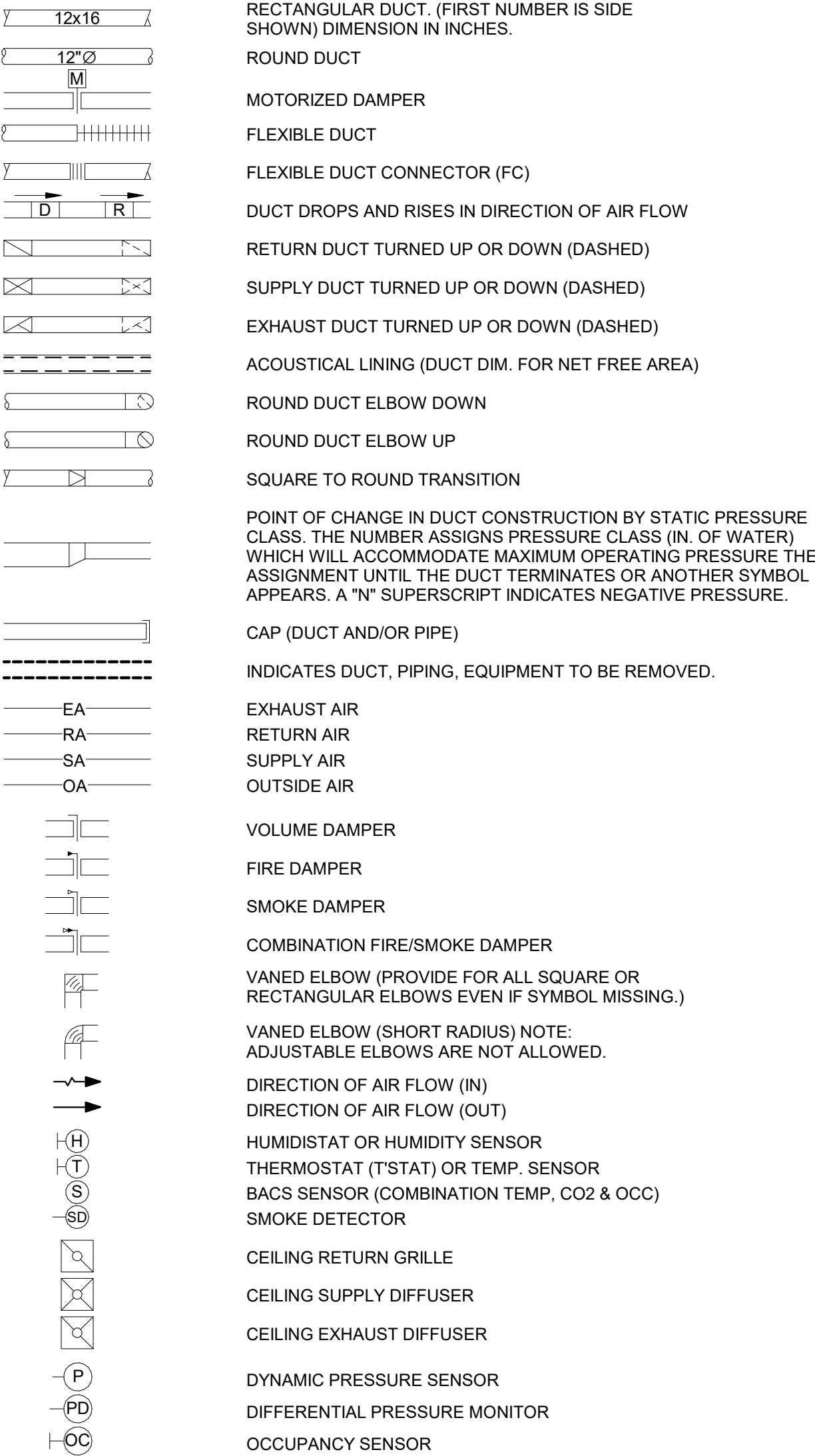
ENGINEERING | ENVIRONMENTAL | SURVEYING

Autodesk Civil/10377.041 - 1-EGS - Jonesboro B&W Heat Pump Upgrades/10377.041 - Jonesboro Heat Pump_MEP.dwg

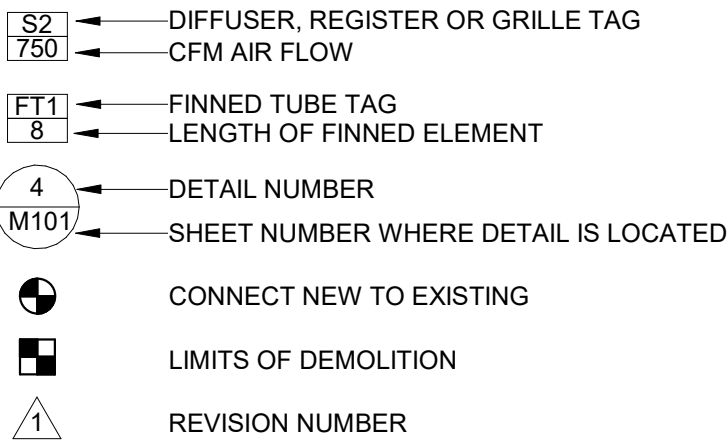
ABBREVIATIONS	
@	AT
A	AMP
ABS	ACRYLONITRILE BUTADIENE STYRENE PLASTIC
ABV	ABOVE
AD	ACCESS DOOR
ADA	AMERICANS WITH DISABILITIES ACT
AF	AIR FILTER
AFF	ABOVE FINISHED FLOOR
AFM	AIR FLOW MEASURING STATION
AL	ACOUSTICAL LINER
AMB	AMBIENT
AP	ACCESS PANEL
APD	AIR PRESSURE DROP
APPROX	APPROXIMATELY
ASME	AMERICAN SOCIETY OF MECHANICAL ENGINEERS
ASSY	ASSEMBLY
ATC	AUTOMATIC TEMPERATURE CONTROL
ATT	ACOUSTICAL ATTENUATOR
AV	AUTOMATIC VENT
BDD	BACKDRAFT DAMPER
BHP	BRAKE HORSEPOWER
BLDG	BUILDING
BOT	BOTTOM
BOD	BOTTOM OF DUCT
BTUH	BTU PER HOUR
C	CENTERLINE
CA	COMPRESSED AIR
CAP	CAPACITY
CC	COOLING COIL
CFM	CUBIC FEET PER MINUTE
CLS	CEILING
CO	CLEAN OUT, CARBON MONOXIDE
COL	COLUMN
CONC	CONCRETE
CHWR	CHILLED WATER RETURN
CHWS	CHILLED WATER SUPPLY
COND	CONDENSATE
CONN	CONNECTION
CONT	CONTINUATION
CPT	CONTROL PANEL, CONDENSATE PUMP
CV	CONDENSATE PUMP TRAP
Cv	CONTROL VALVE
CvT	CONTROL VALVE RATING
CVT	CONSTANT VOLUME AIR TERMINAL
CW	DOMESTIC COLD WATER
CWR	COLD WATER RETURN
CWS	COLD WATER SUPPLY
D	DRAIN, DEPTH
dB	DECIBELS
DB	DRY BULB
DDC	DIRECT DIGITAL CONTROL
Ø, DIA	DIAMETERS
DIFF	DIFFERENTIAL, DIFFUSER
DISCH	DISCHARGE
DN	DOWN
DOM	DOMESTIC
DP	DIFFERENTIAL PRESSURE
DWG	DRAWING
DWGS	DRAWINGS
EA	EXHAUST AIR
EAT	ENTERING AIR TEMPERATURE
EC	ELECTRICAL CONTRACTOR
EDR	EQUIVALENT DIRECT RADIATION
EER	ENERGY EFFICIENT RATIO
EFF	EFFICIENCY
ELEC	ELECTRIC, ELECTRICAL
ELEV	ELEVATION
ENT	ENTERING
EPDM	ETHYLENE PROPYLENE DIENE MEMBRANE
EQUIP	EQUIPMENT
ESP	EXTERNAL STATIC PRESSURE
EXP	EXPANSION
F	FAN, DEGREES FAHRENHEIT
FA	FRESH AIR
FAI	FRESH AIR INTAKE
FBG	FURNISHED BY GOVERNMENT
FC	FLEX CONNECTION
FCO	FLOOR CLEANOUT
FD	FIRE DAMPER
FF	FINISH FLOOR
FIX	FIXTURE
FLA	FULL LOAD AMPS
FLR	FLOOR
FOB	FLAT ON BOTTOM
FOT	FLAT ON TOP
FS	FLOAT SWITCH
FSD	FIRE SMOKE DAMPER
FTR	FIN TUBE RADIATION
FZ	FREEZE/STAT
G	GAS
GA	GAUGE
GAL	GALLONS
GALV	GALVANIZED
GC	GENERAL CONTRACTOR
GP	GENERAL PURPOSE
GPH	GALLONS PER HOUR
GPM	GALLONS PER MINUTE
GRH	GRAVITY RELIEF HOOD
GV	GATE VALVE
GSM	GALVANIZED SHEET METAL
GYP	GYPNUM WALLBOARD
H	HEIGHT
HOA	HANDS-OFF-AUTOMATIC
HOR	HORIZONTAL
HP	HORSEPOWER, HIGH PRESSURE
HPC	HIGH PRESSURE CONDENSATE
HPS	HIGH PRESSURE STEAM
HR	HOUR
HT	HEIGHT
HUMID	HUMIDIFIER, HUMIDITY
HVAC	HEATING, VENTILATING AND AIR CONDITIONING
HW	DOMESTIC HOT WATER
HWR	HOT WATER RETURN
HWS	HOT WATER SUPPLY
HZ	HERTZ

I/O	INPUT / OUTPUT
ID	INSIDE DIAMETER
IN	INCHES
KW	KILOWATT
L	LENGTH
LAT	LEAVING AIR TEMPERATURE
LD	LIQUEFIED PETROLEUM
LDB	LEAVING DRY BULB
LF	LINEAR FEET
LONG	LONG
LOC	LOCATION, LOCATED
LP	LIQUEFIED PROPANE
LPC	LOW PRESSURE CONDENSATE
LPS	LOW PRESSURE STEAM
LRA	LOCKED ROTOR AMPS
LS	LINE SET (REFRIG)
LVG	LEAVING
LWB	LEAVING WET BULB
LWT	LEAVING WATER TEMPERATURE
MAX	MAXIMUM
MAX PD	MAXIMUM PRESSURE DROP
MBH	1000 BTU PER HOUR
MBU	1000 BTU
MC	MECHANICAL CONTRACTOR
MCA	MAXIMUM CIRCUIT AMPS
MCC	MOTOR CONTROL CENTER
MECH	MECHANICAL
MEZZ	MEZZANINE
MFG	MANUFACTURER
MIN	MINIMUM, MINUTES
MLS	MAIN LINE SET (REFRIG)
mm	MILLIMETERS
MNTD	MOUNTED
MOCP	MAXIMUM OVERCURRENT PROTECTION
MUW	MAKE-UP WATER
N/A	NOT APPLICABLE
NATL	NATURAL
NC	NORMALLY CLOSED, NOISE CRITERIA
NEC	NATIONAL ELECTRIC CODE NFPA 70
NG	NATURAL GAS
NFPA	NATIONAL FIRE PROTECTION ASSOCIATION
NIC	NOT IN CONTRACT
NO	NORMALLY OPEN, NUMBER
N02	NITROGEN DIOXIDE
NTS	NOT TO SCALE
OA	OUTSIDE AIR
OAF	OUTSIDE AIR FILTER, OUTSIDE AIR FAN
OAI	OUTSIDE AIR INTAKE
OAT	OUTSIDE AIR TEMPERATURE
OBVD	OPPOSED BLADE VOLUME DAMPER
OD	OUTSIDE DIAMETER
OED	OPEN ENDED DUCT
OS&Y	OUTSIDE STEM AND YOKE
P	PUMP, PITCH
PC	PUMPED CONDENSATE
PD	PRESSURE DROP
PH	PHASE
PLMB	PLUMBING
PRESS	PRESSURE
PRV	PRESSURE REDUCING VALVE
PSI	POUNDS PER SQUARE INCH
PSIG	POUNDS PER SQUARE INCH GAUGE
PT	PRESSURE TREATED
PVC	POLYVINYL CHLORIDE
QTY	QUANTITY
R	RADIUS, RETURN
RA	RETURN AIR
RAD	RADIATOR
RAF	RETURN AIR FAN
RAT	RETURN AIR TEMPERATURE
REL	RELIEF
REQD	REQUIRED
RET	RETURN
RH	RELATIVE HUMIDITY
RLA	RATED LOAD AMPS
RL	REFRIGERANT LIQUID
ROOM	ROOM
RPM	REVOLUTIONS PER MINUTE
RS	REFRIGERANT SUCTION
SA	SUPPLY AIR
SCH	SCHEDULE
SCR	SCREEN
SD	SMOKE DAMPER
SF	SQUARE FOOT
SIM	SIMILAR
SMACNA	SHEET METAL AND AIR CONDITIONING CONTRACTOR'S NATIONAL ASSOCIATION
SOV	SHUT OFF VALVE
SP	STATIC PRESSURE
SPH	STATIC PRESSURE HIGH LIMIT
SPL	STATIC PRESSURE LOW LIMIT
SPS	STATIC PRESSURE SENSOR
SQ	SQUARE
SS	STAINLESS STEEL
STL	STEEL
SUP	SUPPLY
T	TEMPERATURE SENSOR, THERMOSTAT
TC	TOTAL COOLING
TEMP	TEMPERATURE
THK	THICK, THICKNESS
TG	TRANSFER GRILLE
TRANS	TRANSITION
TSP	TOTAL STATIC PRESSURE
TSTAT	THERMOSTAT
TYP	TYPICAL
UNO	UNLESS NOTED OTHERWISE
V	VENT, VOLT
VD	VOLUME DAMPER
VEL	VELOCITY
VFD	VARIABLE FREQUENCY DRIVE
VIF	VERIFY IN FIELD
VRF	VARIABLE REFRIGERANT FLOW
VTR	VENT THROUGH ROOF
W	WIDTH, WATT
W/	WITH
WB	WET BULB
WC	WATER COLUMN
WCO	WALL CLEANOUT
WG	WATER GAUGE
WMS	WIRE MESH SCREEN
WPD	WATER PRESSURE DROP

AIR DISTRIBUTION SYMBOLS



MISCELLANEOUS SYMBOLS



MECHANICAL EQUIPMENT ABBREVIATIONS

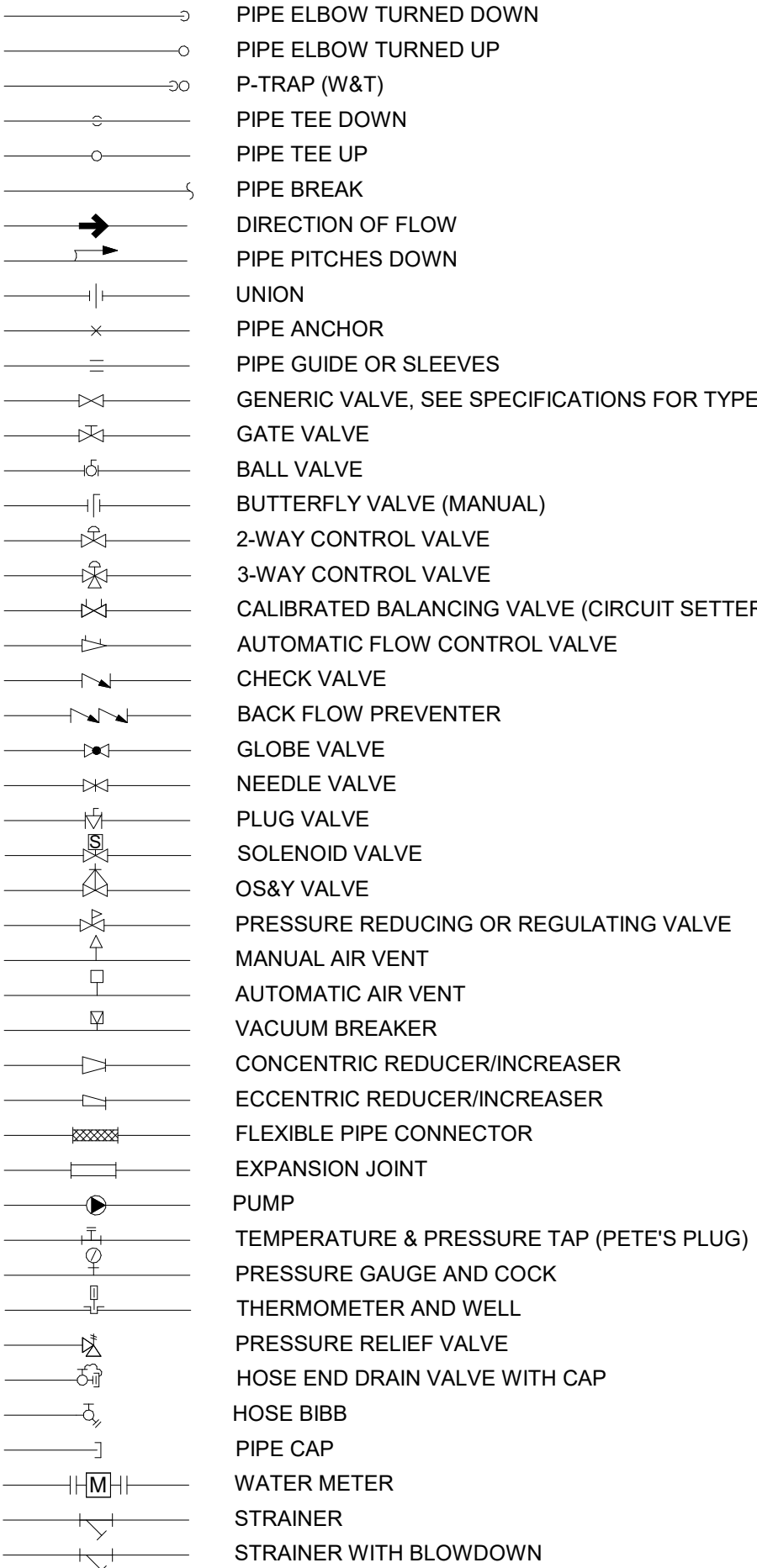
(REFER TO MECHANICAL SCHEDULE SHEET FOR INFORMATION)

AHU	AIR HANDLING UNIT
EF	EXHAUST FAN
ERV	ENERGY RECOVER VENTILATOR
EBH	ELECTRIC BASEBOARD HEATER
HP	HEAT PUMP
DHC	DUCT HEATING COIL
DBE	DRYER BOOSTER FAN
IU	HEAT PUMP INDOOR UNIT
OU	HEAT PUMP OUTDOOR UNIT
UH	UNIT HEATER
ET	EXPANSION TANK
VAV	VARIABLE AIR VOLUME

EXISTING EQUIPMENT LEGEND

(E)	EXISTING TO REMAIN
(R)	EXISTING TO BE DISCONNECTED AND REMOVED
(RL)	EXISTING TO BE DISCONNECTED AND RELOCATED
(ER)	EXISTING IN NEW LOCATION
(RP)	EXISTING TO BE REPLACED

PIPING SYMBOLS



PIPING SYSTEMS

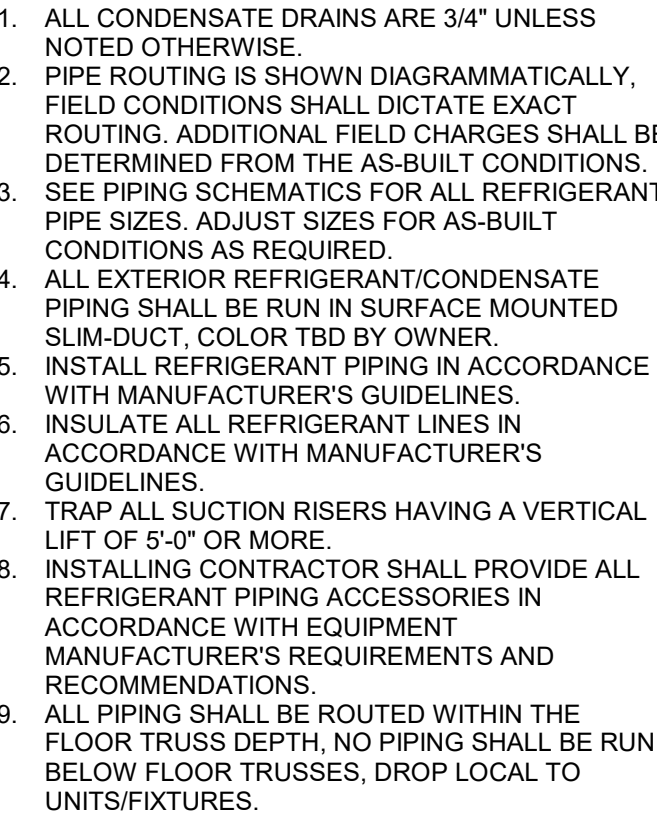
(E)	EXISTING PIPING TO REMAIN
----- (R) -----	EXISTING PIPING TO BE REMOVED
----- HWR -----	HOT WATER RETURN
----- HWS -----	HOT WATER SUPPLY
----- RL/RS -----	REFRIGERANT LIQUID / REFRIGERANT SUCTION
----- RL -----	REFRIGERANT LIQUID
----- RS -----	REFRIGERANT SUCTION
----- RH -----	REFRIGERANT HOT GAS
----- CHWR -----	CHILLED WATER RETURN
----- CHWS -----	CHILLED WATER SUPPLY
----- CD -----	CONDENSATE DRAIN
----- CA -----	COMPRESSED AIR
----- AD -----	ACID DRAIN
----- AD -----	ACID DRAIN (BELOW SLAB)
----- AV -----	ACID VENT
----- CWS -----	CONDENSER WATER SUPPLY
----- CWR -----	CONDENSER WATER RETURN
----- FOR -----	FUEL OIL RETURN
----- FOS -----	FUEL OIL SUPPLY
----- FOV -----	FUEL OIL VENT
----- GYL -----	GLYCOL
----- GYLS -----	GLYCOL SUPPLY
----- GYLR -----	GLYCOL RETURN
----- HCR -----	HIGH PRESSURE CONDENSATE RETURN
----- HLP -----	HIGH PRESSURE LIQUID PROPANE
----- HNG -----	HIGH PRESSURE NATURAL GAS
----- HPS -----	HIGH PRESSURE STEAM
----- LPS -----	LOW PRESSURE STEAM
----- LPR -----	LOW PRESSURE RETURN
----- MPS -----	MEDIUM PRESSURE STEAM
----- MPR -----	MEDIUM PRESSURE RETURN
----- NPW -----	NON POTABLE COLD WATER
----- PCWS -----	PROCESS COOLING WATER SUPPLY
----- PCWR -----	PROCESS COOLING WATER SUPPLY
----- PC -----	PUMP STEAM CONDENSATE
----- PD -----	PUMP DISCHARGE
----- PW -----	POTABLE WATER
----- R -----	RELIEF LINE

GENERAL MECHANICAL NOTES

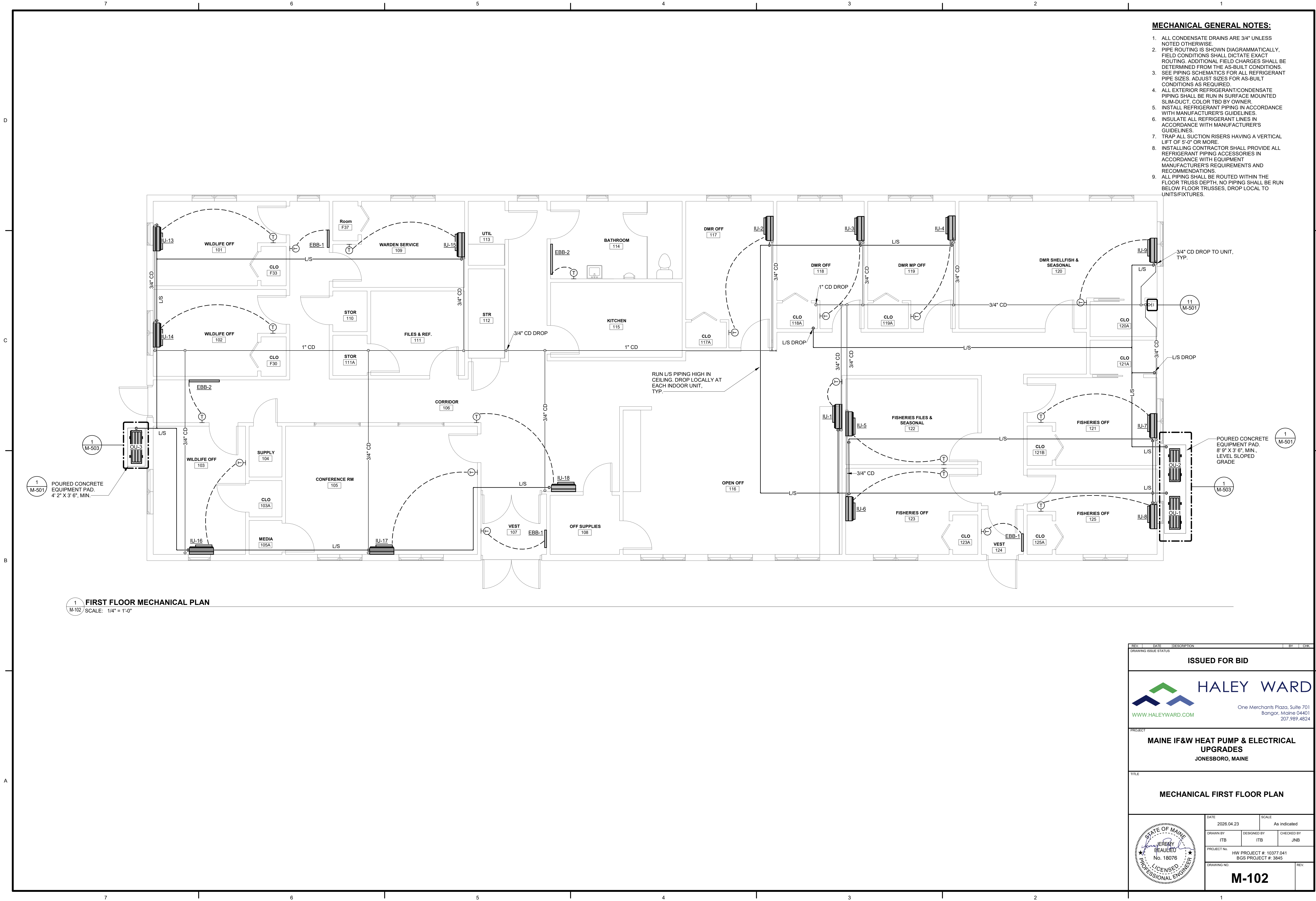
- IT IS THE INTENT OF THESE DRAWINGS TO SHOW COMPLETE AND FUNCTIONAL SYSTEMS THAT ARE IN COMPLIANCE WITH ALL INDUSTRY STANDARDS AND APPLICABLE CODES. ANY DISCREPANCIES SHOULD BE BROUGHT TO THE ENGINEER'S ATTENTION.
- ALL MECHANICAL SYSTEMS WORK SHALL BE IN ACCORDANCE WITH ALL FEDERAL, STATE AND LOCAL AUTHORITIES HAVING JURISDICTION AND ALL APPLICABLE CODES. ALL WORK SHALL CONFORM TO EQUIPMENT MANUFACTURER'S INSTRUCTIONS AND INDUSTRY STANDARDS.
- ALL WORKMANSHIP SHALL BE OF THE HIGHEST STANDARDS. INSTALL ALL WORK IN A NEAT, SYSTEMATIC AND ORDERLY ARRANGEMENT. ALL MATERIAL SHALL BE NEW AND OF THE BEST QUALITY AVAILABLE; FREE FROM DEFECTS. THE CONTRACTOR SHALL GUARANTEE THE MATERIALS AND INSTALLATION FOR ONE YEAR FROM THE PROJECT ACCEPTANCE DATE AGAINST ANY DEFECTS DUE TO FAULTY MATERIALS, EQUIPMENT, WORKMANSHIP, OR INSTALLATION. UPON NOTICE OF THE DEFECT, THE CONTRACTOR SHALL REPLACE OR REPAIR THE DEFECTIVE ITEM AT NO ADDITIONAL COST.
- THE CONTRACTOR SHALL VISIT THE JOB SITE TO VERIFY ALL EXISTING FIELD CONDITIONS. DIMENSIONS AND OBSTRUCTIONS.
- ALL PIPING AND DUCTWORK IS SHOWN DIAGRAMMATICALLY. PIPING AND SYSTEMS SHALL FOLLOW ARRANGEMENT AS MUCH AS POSSIBLE, HOWEVER, ACTUAL FIELD CONDITIONS SHALL DICTATE. CAREFULLY COORDINATE THE SPACE REQUIREMENTS AND LOCATIONS OF ALL DUCTWORK WITH ALL OTHER TRADES. GIVE PRIORITY TO GRAVITY DRAINAGE PIPING.
- THE CONTRACTOR SHALL COORDINATE ALL WORK WITH THE WORK OF ALL OTHER TRADES. SHOULD ANY DISCREPANCIES BE DISCOVERED IN ANY OF THE BID DOCUMENTS, (INCLUDING ALL OTHER DIVISIONS) THE CONTRACTOR SHALL IMMEDIATELY NOTIFY THE ENGINEER BEFORE PROCEEDING ANY FURTHER WITH THE WORK, OTHERWISE THE CONTRACTOR SHALL BE HELD RESPONSIBLE FOR ALL COST INVOLVED WITH THE CORRECTION OF THE CONSTRUCTION INSTALLATION.
- ALL EQUIPMENT, DUCTWORK, PIPING, ETC. SHALL BE SUPPORTED FROM THE BUILDING'S STRUCTURAL FRAME AND MEMBERS. ALL DUCT SIZES ARE NET DIMENSIONS AND DO NOT INCLUDE AND INSULATION. SUPPORT OR REINFORCEMENT DIMENSIONS. ALL WORK SHALL BE NEW UNLESS OTHERWISE NOTED AS EXISTING.
- THE CONTRACTOR SHALL PERFORM TESTS ON ALL MECHANICAL SYSTEMS AS REQUIRED BY FEDERAL, STATE, AND LOCAL CODES AND REGULATIONS. ALL TESTS SHALL BE WITNESSED AND ACCEPTED BY THE AUTHORITY HAVING JURISDICTION. THE CONTRACTOR SHALL PROVIDE ALL SERVICES AND MATERIALS REQUIRED BY THE TEST AND CERTIFY IN WRITING THAT ALL WORK HAS PASSED ALL REQUIRED TESTS.
- THE MECHANICAL SYSTEMS SHALL BE BALANCED, COMPLETE WITH A WRITTEN REPORT BY AN INDEPENDENT AIR BALANCE FIRM WITH A MINIMUM OF 3 YEARS EXPERIENCE.
- WHERE PIPES AND DUCTS PENETRATE WALLS OR FLOOR, THE SPACE BETWEEN THE STRUCTURE AND THE DUCT OR PIPE SHALL BE SEALED AIRTIGHT WITH AN APPROVED MATERIAL. PROVIDE FIRE STOPS AND/OR SEALANT AROUND ALL PENETRATIONS THAT HAVE A FIRE RATING GREATER THAN OR EQUAL TO THE FIRE RATING OF THE WALL, FLOOR OR ENCLOSURE.
- PROVIDE ACCESS PANELS FOR ALL VALVES, DAMPERS, CLEANOUTS, ETC. THAT REQUIRE ACCESS.

SHEET LIST - MECHANICAL	
M-001	MECHANICAL NOTES, SYMBOLS & ABBREVIATIONS
M-101	MECHANICAL BASEMENT PLAN
M-102	MECHANICAL FIRST FLOOR PLAN
M-501	MECHANICAL DETAILS
M-502	MECHANICAL DETAILS
M-503	MECHANICAL DETAILS
M-601	MECHANICAL SCHEDULES

REV	DATE	DESCRIPTION	BY	CHK
DRAWING ISSUE STATUS				
ISSUED FOR BID				
 HALEY WARD One Merchants Plaza, Suite 701 Bangor, Maine 04401 207.989.4824 WWW.HALEYWARD.COM				
PROJECT MAINE IF&W HEAT PUMP & ELECTRICAL UPGRADES JONESBORO, MAINE				
TITLE MECHANICAL NOTES, SYMBOLS & ABBREVIATIONS				
DATE 2026.04.23		SCALE 12" = 1'-0"		
DRAWN BY ITB	DESIGNED BY ITB	CHECKED BY JNB		
PROJECT No. HW PROJECT #: 10377.041 BGS PROJECT #: 3845				
DRAWING NO. M-001		REV		



DATE 2026.04.23		SCALE As indicated	
DRAWN BY ITB	DESIGNED BY ITB	CHECKED BY JNB	
PROJECT No. HW PROJECT #: 10377.041 BGS PROJECT #: 3845			
DRAWING NO. M-101			REV.



MECHANICAL GENERAL NOTES:

1. ALL CONDENSATE DRAINS ARE 3/4" UNLESS NOTED OTHERWISE.
2. PIPE ROUTING IS SHOWN DIAGRAMMATICALLY, FIELD CONDITIONS SHALL DICTATE EXACT ROUTING. ADDITIONAL FIELD CHARGES SHALL BE DETERMINED FROM THE AS-BUILT CONDITIONS.
3. SEE PIPING SCHEMATICS FOR ALL REFRIGERANT PIPE SIZES. ADJUST SIZES FOR AS-BUILT CONDITIONS AS REQUIRED.
4. ALL EXTERIOR REFRIGERANT/CONDENSATE PIPING SHALL BE RUN IN SURFACE MOUNTED SLIM-DUCT, COLOR TBD BY OWNER.
5. INSTALL REFRIGERANT PIPING IN ACCORDANCE WITH MANUFACTURER'S GUIDELINES.
6. INSULATE ALL REFRIGERANT LINES IN ACCORDANCE WITH MANUFACTURER'S GUIDELINES.
7. TRAP ALL SUCTION RISERS HAVING A VERTICAL LIFT OF 5'-0" OR MORE.
8. INSTALLING CONTRACTOR SHALL PROVIDE ALL REFRIGERANT PIPING ACCESSORIES IN ACCORDANCE WITH EQUIPMENT MANUFACTURER'S REQUIREMENTS AND RECOMMENDATIONS.
9. ALL PIPING SHALL BE ROUTED WITHIN THE FLOOR TRUSS DEPTH, NO PIPING SHALL BE RUN BELOW FLOOR TRUSSES, DROP LOCAL TO UNITS/FIXTURES.

1 FIRST FLOOR MECHANICAL PLAN
M-102 SCALE: 1/4" = 1'-0"

REV.	DATE	DESCRIPTION	BY	CHK.
DRAWING ISSUE STATUS				
ISSUED FOR BID				
 HALEY WARD One Merchants Plaza, Suite 701 Bangor, Maine 04401 207.989.4824 WWW.HALEYWARD.COM				
PROJECT MAINE IF&W HEAT PUMP & ELECTRICAL UPGRADES JONESBORO, MAINE				
TITLE MECHANICAL FIRST FLOOR PLAN				
DATE 2026.04.23		SCALE As indicated		
DRAWN BY ITB		DESIGNED BY ITB		CHECKED BY JNB
PROJECT No. HW PROJECT #: 10377.041 BGS PROJECT #: 3845				
DRAWING NO. M-102		REV.		

